SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

<u>Descriptive Written Test</u>

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

Aim

To study intelligent agents in a smart home automation system and identify the most suitable agent architecture.

Objectives

• Understand intelligent agents.
• Analyze Simple Reflex, Model-Based, Goal-Based and Utility-Based agents.
• Compare their applications.
• Recommend the best approach.

Introduction

A smart home automation system integrates AI to automatically control lighting, temperature and security. The system continuously senses environmental conditions and user behaviour to improve comfort, energy efficiency and safety.

Problem Understanding

The system receives real-time inputs such as motion detection, room temperature and time of day. It must respond intelligently while learning user preferences over time.

Explanation

Simple Reflex Agent reacts only to current input using condition-action rules.

Model-Based Agent stores information about previous states and predicts future states.

Goal-Based Agent selects actions that achieve goals such as maintaining room temperature or securing the house.

Utility-Based Agent evaluates multiple possible actions and chooses the one providing maximum overall benefit considering comfort, security and energy consumption.

Application

Simple Reflex can switch ON lights when motion is detected.
Model-Based remembers occupancy patterns.
Goal-Based ensures desired temperature.
Utility-Based balances comfort with electricity consumption.
A Hybrid Agent combining Model-Based and Utility-Based approaches is the most effective because it learns user behaviour while optimizing comfort, energy efficiency and security.

Justification

Hybrid architecture provides adaptability, intelligence, lower power consumption and better decision making than individual agent types.

Conclusion

A hybrid Model-Based + Utility-Based intelligent agent offers the best overall performance for smart home automation.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Aim

To analyze how UCS and IDS solve an unknown maze and recommend the most suitable intelligent agent.

Objectives

• Study UCS and IDS.
• Compare time and space complexity.
• Relate search algorithms to intelligent agents.
• Recommend the best solution.

Problem Understanding

The robot starts without any knowledge of the maze and must search until it finds the exit while minimizing cost.

Uniform Cost Search

UCS expands the node with the lowest cumulative path cost. It always finds the optimal path when costs are positive. Time and space complexity are exponential in worst case.

Iterative Deepening Search

IDS performs repeated depth-limited searches until the goal is found. It combines DFS memory efficiency with BFS completeness. Time complexity $O(b^d)$; Space complexity $O(bd)$.

Advantages

UCS: Optimal, complete.
IDS: Low memory, complete.

Disadvantages

UCS: High memory and computational cost.
IDS: Repeated expansions increase execution time.

Agent Relation

A Goal-Based Agent searches until it reaches the exit. A Utility-Based Agent additionally considers energy consumption and safety while selecting paths.

Recommended Solution

A Utility-Based Goal-Oriented Agent using Uniform Cost Search is recommended because it minimizes total travel cost while guaranteeing the optimal path.

Conclusion

Combining Utility-Based intelligence with UCS provides safe, efficient and optimal maze navigation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand